



Conditional Motif-Based Graph Convolutional Network for Anomaly Detection in the Waste Management Network

Sara Oliveira¹(✉) , Shazia Tabassum¹ , João Gama¹ , Ana Garcia²,
and Pedro Santana²

¹ INESC TEC Porto, 4200-465 Porto, Portugal

sara.l.oliveira@inesctec.pt

² IGAMAOT, 1200-433 Lisboa, Portugal

<https://www.inesctec.pt>, <https://www.igamaot.gov.pt>

Abstract. Illicit activities in the waste management network, such as waste laundering, misreporting, or trade of stolen waste pose serious environmental and regulatory challenges. Detecting these behaviours is challenging, because they often emerge from higher-order interactions among multiple entities, and are not continuous over time. Furthermore, these activities often manifest as triangles in the network, and the participation of individuals in these waste transfer structures is additionally suspicious. Traditional anomaly detection methods, which rely on first-order relationships or static analyses, struggle to capture these complex, temporally dynamic patterns. To address this challenge, we propose a Conditional Motif-Based Graph Convolutional Network (CM-GCN) that integrates condition-driven triangular motifs directly into the GCN message-passing mechanism. The CM-GCN learns structural embeddings that encode both local graph topology and node attributes-based connectivity to triangular motifs. To detect sudden or sporadic changes, these weekly embeddings are processed by a Long Short-Term Memory Variational Autoencoder (LSTM-VAE), which models temporal behaviour and identifies anomalies through spikes in reconstruction error. Experiments on one year of Portuguese waste transport data demonstrate that the proposed approach effectively highlights companies with known illicit behaviour. The CM-GCN-LSTM-VAE outperformed a standard GCN-LSTM-VAE that ignores motif structure. Results are comparable to, and slightly improve upon, an LSTM-VAE trained on a manually engineered triangle-based feature. This demonstrates that higher-order structural representations learned by the model provide a more informative signal, while simple pairwise relationships contribute little to the detection of complex behaviours.

Keywords: Graph Convolutional Network · Autoencoder · Waste Management

1 Introduction

Waste transportation plays a vital role in the broader waste management system. Waste produced by industrial, commercial, and household activities must be transferred to facilities where they can be treated, recycled, or disposed of in accordance with environmental standards. The proper functioning of this system has direct implications for environmental protection, resource efficiency, and public health. The management of waste transportation in Portugal is governed by strict environmental regulations which are designed to minimize pollution, prevent illegal dumping and trade of stolen waste, and overall promote accountability among the entities that are involved in the waste chain.

Despite this structured regulatory environment, waste transport monitoring remains a challenging task. Waste management operations involve numerous stakeholders, producers, transporters, intermediaries, and recipients. Some entities may engage in falsification, misreporting, or omission of waste transport data to circumvent regulatory obligations, reduce costs, or avoid taxes associated with proper waste disposal. In more serious cases, these actions develop into organized crimes involving the theft and illegal trade of valuable materials such as scrap metal or car parts. Others take advantage of regulatory gaps between countries, engaging in cross-border waste trafficking that shifts environmental burdens to less regulated destinations.

To address these risks, Portuguese authority IGAMAOT (The General Inspectorate of Agriculture, Sea, Environment and Spatial Planning) [9], carries out inspections and audits to detect irregularities and enforce compliance. However, the volume of data generated through e-GAR [7], the mandatory electronic documentation system for waste shipments, is huge. This makes manual analysis increasingly impractical.

Thus, an efficient, data-driven tool is needed to detect waste operating companies engaging in potentially illicit activities, to be flagged for inspection. However, these anomalies can be difficult for traditional fraud detection methods to detect, given that they often emerge from the collusive relation of several entities in the network and that they typically happen sporadically, among periods of normal activity. Thus, they require an approach that can capture the higher-order network topology and its temporal dynamics. Additionally, illicit activities in the waste management network have been confirmed by the domain experts to frequently manifest as triangle formations, and individual waste senders are associated with informal waste trades and thus suggestive of suspicious activities.

Therefore, we propose a time-aware anomaly detection framework with a novel conditional motif-based Graph Convolutional Network (CM-GCN). CM-GCN integrates condition-driven triangular motifs into its message-passing mechanism. By conditioning propagation on the presence of specific motif patterns, CM-GCN embeds higher-order connectivity patterns directly into node representations. Across time, the CM-GCN produces conditional motif-informed dynamic embedding timeseries for each node, which are then processed by a Long Short-Term Memory Variational Autoencoder (LSTM-VAE) to detect temporal anomalies through periods of high reconstruction error. This approach is com-

pared to a standard GCN–LSTM–VAE model, and to a LSTM–VAE model that replaces graph embeddings with manually engineered graph theoretical time-series feature as input.

2 Literature Review

Graph anomaly detection identifies unusual patterns in network data like nodes, edges, or subgraphs that deviate from normal behavior, with applications in fraud detection, cybersecurity, and social networks. Graph Neural Networks (GNNs) have seen growing research interest, due to their ability to learn graph representations through their message passing mechanism, that aggregates the feature information from the neighbours of a given node [10]. Therefore, they have been used for several tasks, including anomaly detection. Graph Convolutional Networks (GCNs) are one of the most widely used for anomaly detection, particularly in unsupervised settings [14]. They learn graph data by aggregating and transforming node features through multiple layers of convolutions, similar to traditional convolutional networks but adapted for irregular graph structures [5, 13, 22]. Other notable GNN architectures include Graph Attention Networks (GATs) [6, 21], which use attention mechanisms to learn different levels of importance of the neighbors when aggregating information; and GraphSAGE [4, 8], which uses neighbour sampling and learns flexible neighborhood aggregation functions. While GNNs effectively learn meaningful graph representations, they are typically combined with other deep learning architectures, to capture other types of information, such as temporal patterns [10]. These additional architectures include Transformers [13], Gated Recurring Units [22] and Long Short–Term Memory networks [18].

Traditional GNN–based anomaly detection methods typically rely on pairwise relationships between first–order neighbors, and are limited in expressiveness to what the 1–dimensional Weisfeiler–Lehman (1–WL) graph isomorphism test can distinguish [15]. Therefore, these standard GNN architectures may be unable to capture anomalies associated with complex higher–order structures. This has lead to the development of motif–aware approaches. Motifs refer to any recurrent subgraph patterns, such as wedges, triangles, or more complex structures, that represent interaction patterns beyond simple edges. Instead of aggregating information solely from immediate neighbors, these approaches embed motif–induced adjacency matrices [11, 12] or aggregate information from motif–derived “supernodes” [19, 20] into the GNN architecture, to enhance node representations and node–level anomaly detection, or to detect anomalous motif instances. However, the dependency of motifs on specific contextual conditions or attribute–driven categories have not been explored until now. Furthermore, integrating such conditional motif reasoning within a sequential, time–aware framework for dynamic graph anomaly detection introduces an additional layer of novelty, enabling the identification of evolving structural irregularities over time. Other approaches have also been developed to capture higher–order structures in GNNs. k –WL based approaches [1, 15] aim to overcome the expressive

limitations of standard GNNs by aligning architectures with the k -WL hierarchy of graph isomorphism tests. These k -dimensional GNNs (k -GNNs) run message passing through higher-order graph structures (k -tuples of nodes). While theoretically powerful, these methods face significant computational overhead, limiting applicability. Message Passing Simplicial Networks (MPSNs) [3] instead perform message passing on simplicial complexes. CW networks [2] generalize this idea further through more flexible cell complexes, allowing for multi-dimensional message passing on a broader class of higher-order cells, rather than focusing specifically on predefined motifs.

In the context of waste management, literature on anomaly detection remains sparse. A notable exception [17] proposed a graph-based framework for analyzing the e-GAR waste transportation data. The authors combine network-based analysis and machine learning to detect anomalies in the waste transportation data. Their method integrates sub-graph detection, community structure analysis, and centrality measures to extract structural features, and machine learning techniques, including clustering, autoencoders, and density-based methods, are then applied for detecting anomalies. Our previous work [16] extended this approach by introducing temporal analysis through change detection methods, the Page-Hinkley and CUSUM tests, and deep learning, through a LSTM-VAE model, on features created using Social Network Analysis. The results suggested that the feature representing triangles with certain node types provided the strongest signal of anomalous activity. In this work, we will further build on this approach, leveraging the informative power of attribute-based triangle motifs suggested by the results obtained in [16], with the ability of Graph Neural Networks to learn meaningful representations of connectivity patterns.

3 Dataset

The dataset analyzed in this study consists of a comprehensive collection of anonymized electronic Waste Transport records (e-GAR), documenting the movement of waste materials across Portugal during the year 2022. Initially, the raw data comprised over 4.5 million transport records, which underwent extensive preprocessing and cleaning to remove inconsistencies, incomplete entries, and redundant information. After this process, the final dataset contained approximately 3.5 million valid transfer records. These correspond to transactions involving more than 111,000 unique waste producers (senders) and 1,591 distinct authorized receiver companies (which may also send waste themselves). Each record includes details about the sender and receiver, the type and quantity of waste, timestamp of each operation, the operation codes, type of material and more. While this data is rich in information, it also possesses an inherent network structure from the relationships based on transfers. These relationships are evolving over time.

The dataset is unlabeled without the information on fraudulent activities or users in the network. However, a very small subset of four entities is provided by the client who have been confirmed to have engaged in suspicious or fraudulent activities. Due to the extreme scarcity of these labeled nodes, they are

not used for training of a supervised framework. Rather, they serve as reference points for evaluating model performance in detecting anomalous behavior, and for parameter tuning.

4 Methodology

The adopted methodology consists of a sequential processing framework that processes weekly transaction graphs to detect anomalous company behaviour. After data preprocessing, each weekly graph is encoded using a Conditional Motif-Based GCN, which integrates both pairwise edges and attribute-centered triangle motifs to produce structural embeddings. These embeddings are then passed through an LSTM to model temporal dynamics across weeks, and a variational autoencoder (VAE) reconstructs the temporal embeddings and assigns an anomaly score based on reconstruction error, indicating how strongly a company’s behaviour deviates from learned normal patterns. The overall workflow is illustrated in Fig. 1.

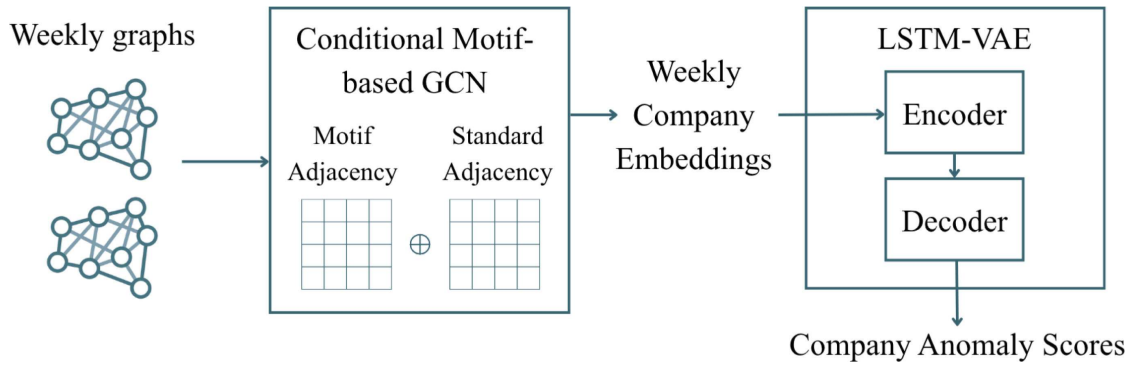


Fig. 1. Overview of the Anomaly Detection model.

4.1 Data Preprocessing

Before conducting our analysis, we performed several data preprocessing steps to ensure the quality of the data. Firstly, we removed all e-GAR documents that had invalid states, such as ‘Nullified’, ‘Rejected’, or ‘Deleted’. Additionally, we removed residual transfers from 2021 and 2023, restricting our analysis to waste transfers taking place in 2022. Operations related to civil construction were also excluded from the dataset, based on guidance from inspectors who indicated that these records were not relevant for the objectives of our study. For the temporal aspects of our analysis, the date corresponding to the beginning of the transport was selected, as it most accurately reflects the actual activity of the waste operators. After these cleaning steps, the resulting dataset contained over 3.5 million transfer records, with more than 111,000 unique senders and

1,591 unique waste receiving companies, which are the target of our anomaly detection.

Using this cleaned data, we constructed weekly graphs composed of the waste transfers taking place between the entities during each given week. The nodes correspond to the senders and receivers of waste, which can be individuals, waste recycling companies or other organizations. The directed edges in the network represent the flow of waste from sender to receiver, and are attributed with the date of transfer. Nodes are further attributed with the type of entity. Specifically, each weekly graph is composed of the target companies and their 2-hop neighbourhoods, i.e., their direct neighbors, and the neighbors of those neighbors.

4.2 Problem Formulation

The dynamic waste-transfer system is modeled as a discrete temporal directed graph comprising heterogeneous node types

$$\mathcal{G} = \{G_1, G_2, \dots, G_T\},$$

where each snapshot $G_t = (V_t, E_t, X_t)$ represents the network at time t . Here, V_t is the set of nodes belonging to different node types/categories (e.g., waste producing companies, transporting companies, individuals, unauthorized entities or waste recycling companies), $E_t \subseteq V_t \times V_t$ denotes the directed waste transfer relations, Each node $v_i \in V_t$ is associated with a categorical feature indicating whether it belongs to a category of interest.

$$x_i^{(t)} = \begin{cases} 1, & \text{if node } i \text{ belongs to the category of interest } C, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, a node belonging to a category of interest C is considered a node-of-interest. In our case, the category of interest is individuals, as they are typically associated with informal waste trading. Note that nodes-of-interest are not pre-labeled fraudulent nodes and are not subject to prior inspection, and their probability of exhibiting fraudulent behaviour is assumed to be the same as any other node in the network. However, the participation of waste receiving companies, the target of anomaly detection, in triangles with individuals is considered highly suspicious by the domain experts, and is therefore the target of our analysis.

In addition, more features from data (for example: type of operation, the type of waste material handled, hazardous materials) can be incorporated into the feature matrix to provide richer contextual information and weighting the conditional representations learned by the model.

4.3 Conditional Motif-Based Graph Convolutional Network

A graph convolutional network (GCN) is a neural network architecture specifically designed to operate on graph-structured data. GCNs extend the notion of convolution from regular grids to irregular domains such as networks. In a traditional GCN, each node updates its representation by aggregating the features of

its neighbors through a propagation function. This allows the network to learn latent embeddings that reflect structural proximity: nodes that are connected or share many neighbors tend to obtain similar representations. However, traditional GCNs rely exclusively on pairwise edges and therefore cannot explicitly capture higher-order connectivity patterns, such as triangles or other network structures. To model such patterns, we present CM-GCN where we augment the GCN with a conditional triangle motif adjacency matrix, aiming to capture irregular activity with nodes-of-interest participating in triangles, which are highly suggestive of suspicious activity.

For computing the motif adjacency matrix, for each pair of nodes (i, j) , we count the number of triangles (i, j, k) where one of the nodes is a node-of-interest (individual waste sender). Each triangle is assigned a weight of 1. The resulting motif adjacency is $M_{ij} = \sum_{k \in \Gamma(i) \cap \Gamma(j)} 1$ if triangle (i, j, k) contains at least one node-of-interest, and 0 otherwise, where $\Gamma(i)$ denotes the neighbors of node i . Therefore, M assigns higher weight to node pairs that jointly participate in many triangles including individual waste senders.

On the other hand, the standard adjacency matrix A encodes the observed pairwise relationships between nodes, that is $A_{ij} = 1$ if there is an edge between i and j , and 0 otherwise. Both A and M are symmetrically normalized with added self-loops using

$$\hat{A} = D_A^{-\frac{1}{2}}(A + I)D_A^{-\frac{1}{2}} \quad \text{and} \quad \hat{M} = D_M^{-\frac{1}{2}}(M + I)D_M^{-\frac{1}{2}},$$

where D_A and D_M denote the corresponding degree matrices.

The matrices are then processed in parallel during training using the ReLU activation function:

$$H_A = \text{ReLU}(\hat{A}XW_A), \quad H_M = \text{ReLU}(\hat{M}XW_M), \quad (1)$$

The final embedding is a combination of the two representations:

$$H = \alpha \odot H_A + (1 - \alpha) \odot H_M, \quad (2)$$

where $\alpha \in (0, 1)$ is a learned coefficient that adaptively determines how much each node should rely on local pairwise structure versus motif-induced higher-order structure.

The model is trained across all weeks and target nodes, excluding the four known anomalous nodes, to minimize a loss that combines the reconstruction loss of the motif adjacency matrix with a temporal smoothing term:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}} = \|HH^\top - \hat{M}\|_2^2 + \lambda_{\text{temp}} \|H_c^{(t)} - H_c^{(t-1)}\|_2^2, \quad (3)$$

where $HH^\top$ is the similarity matrix from the embeddings, and λ_{temp} controls the strength of temporal smoothing. This encourages the model to preserve the structure of node-of-interest centered triangle motifs while maintaining temporal consistency of the embeddings.

After training, the best model is re-run on each weekly graph to extract the embeddings for all target nodes, which are then passed to the LSTM-VAE model.

4.4 LSTM Sequence Autoencoding for Temporal Anomaly Detection

The node embeddings obtained by the Conditional motif-aware GCN are then passed into a Long Short-Term Memory Variational Autoencoder (LSTM-VAE) model to detect anomalies over time. The LSTM-VAE is a deep generative model that combines the ability of LSTM networks to model sequential data with the probabilistic latent variable framework of Variational Autoencoders. The LSTM component captures long-term dependencies and temporal dynamics in the companies' behavior, while the Variational Autoencoder provides the training objective of learning a compressed, probabilistic representation of normal activity that is robust to noise and variability in the data. The model consists of an encoder LSTM that processes input windows of shape (t, n) , where t is the window size and n is the number of features. The encoder outputs two vectors, the mean and the log-variance, which define the approximate posterior distribution over the latent variables. A latent vector is then sampled from this distribution using the reparameterization trick. This sampled latent vector is passed to a decoder LSTM, which reconstructs the original input sequence. The model is trained to minimize a loss function that combines two terms: the reconstruction loss and the Kullback–Leibler divergence between the approximate posterior $q(z | X)$ and a standard normal prior $p(z) = \mathcal{N}(0, I)$, where z denotes the latent variable. The impact of the KL divergence term on the total loss is controlled by the parameter β , which regulates how much weight is given to the KL divergence term relative to the reconstruction loss. The reconstruction loss is defined as the mean squared error between an input sequence X and its reconstruction $\hat{X}$.

$$\mathcal{L} = \|X - \hat{X}\|_2^2 + \beta D_{\text{KL}}(q(z | X) \| p(z)), \quad (4)$$

High reconstruction errors indicate that the observed sequence deviates from the patterns seen during training, and can therefore be used for anomaly detection.

The model was implemented using the `Keras` library in Python, using sliding windows constructed over each company's time series of embeddings. After training, the model achieving the minimum training loss is used to reconstruct the sequences, and the anomaly score for each company is defined as the mean of its 10 highest reconstruction errors. The top 10 errors were selected instead of the mean of all errors because the objective is to find spikes in periods where it was especially difficult for the model to reconstruct the pattern, as the premise of our approach is that suspicious activities happen sporadically rather than continuously. On the other hand, using the mean reconstruction error would dilute the signal from isolated suspicious spikes.

5 Experiments and Results Discussion

The Conditional motif-based GCN-LSTM-VAE model was applied to the full dataset, consisting of 1,591 target companies and 52 weeks of data. The GCN

module used two graph-convolutional layers with hidden dimension 16, producing node embeddings of size 8, and was trained for 15 epochs with learning rate 5×10^{-4} , weight decay 10^{-6} , and no dropout.

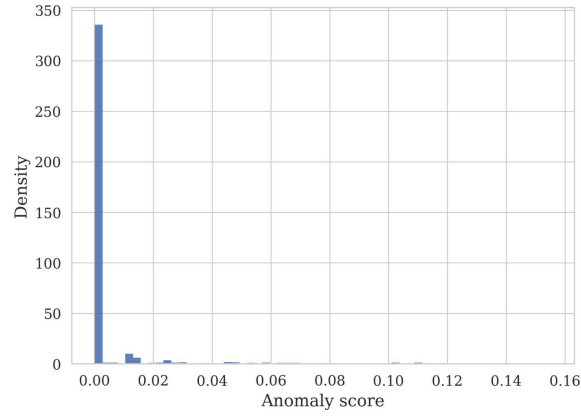


Fig. 2. Distribution of anomaly scores across all companies.

Table 1. Anomaly score percentiles of the four anomalous companies.

Company	Percentile (%)
A	99.50
B	91.95
C	80.26
D	99.56

The weekly embeddings were then arranged into temporal windows of length 8 and fed into the LSTM-VAE for anomaly detection. The encoder contained an LSTM with 32 hidden units followed by two dense layers producing an 8-dimensional latent mean and log-variance. The latent space was regularized through a β -weighted KL divergence with $\beta = 0.5$. The LSTM-VAE was trained for 15 epochs with learning rate 5×10^{-4} , batch size 32, and dropout 0.1. Hyperparameters were tuned by defining ranges for each hyperparameter, testing 15 random combinations, and selecting the combination achieving the best performance on the four anomalous companies.

After running the full pipeline, each target company obtained an anomaly score, reflecting whether it presented periods of unusual connectivity Fig. 2 shows the resulting distribution of anomaly scores.

The obtained distribution is highly positively skewed, with most companies receiving a near-zero anomaly score, consistent with the expectation that most companies exhibit normal behavior. The anomaly score percentiles of the four anonymized anomalous companies are presented in Table 1. Companies A and

D were strongly detected, lying in the top 0.5% of anomaly scores, followed by Company B in the top 8.05%. Company C was weakly detected, appearing in the top 19.74%. Figure 3 shows the static network of three of the pre-labeled anomalous nodes A, B and C only (as the network of all the nodes in the network is infeasible to visualize due to its size). Consistent with the adopted methodology, these nodes jointly participate in triangles including nodes-of-interest (individuals).

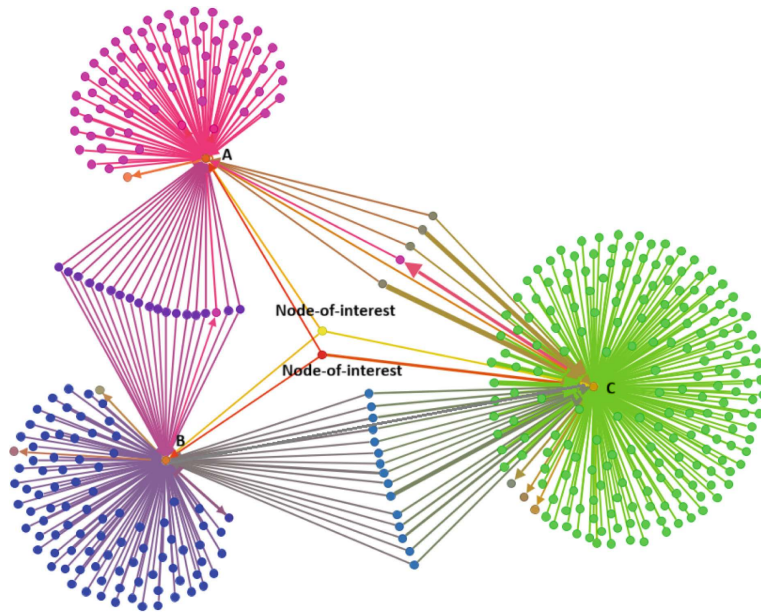


Fig. 3. Network of detected anomalous nodes A, B and C. The node colors in the network are influenced by the color of their neighbors, analogous to message passing in GCN.

5.1 Ablation Study

The performance of the proposed Conditional Motif-based GCN (CM-GCN-LSTM-VAE) model was assessed by comparison against two baseline approaches. The first baseline was a standard GCN-LSTM-VAE relying only on pairwise edges rather than conditional motifs, to detect deviations in companies' overall network connectivity. The second baseline (F-LSTM-VAE) replaced the GCN embeddings with a simple Graph theoretical time-series input feature: for each company and week, the feature consisted of the total number of triangles involving nodes-of-interest connected to that company. An LSTM-VAE was then trained on this sequence to identify anomalous deviations in the company's connectivity patterns. To maintain consistency, all models were trained using the same hyperparameter configuration as the CM-GCN-LSTM-VAE. The resulting anomaly scores for the four anomalous companies under each method are presented in Fig. 4.

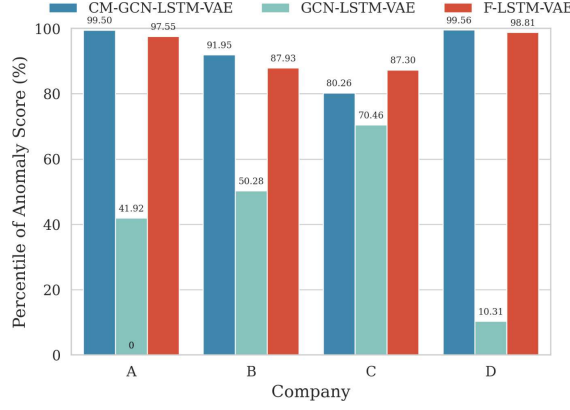


Fig. 4. Comparison of Conditional Motif-based GCN-LSTM-VAE with two baselines.

Firstly, the GCN-LSTM-VAE did not effectively detect any of the four anomalies, assigning them anomaly scores in very low percentiles, which suggests that pairwise connectivity alone is not sufficient to capture anomalous behaviour in the waste management network. Therefore, augmenting the GCN with a category-centered triangle connectivity contributed positively to the detection performance of the GCN-LSTM-VAE model. On the other hand, the F-LSTM-VAE model obtained similar detection performance to the CM-GCN-LSTM-VAE, with the Conditional Motif-based model obtaining a higher anomaly score percentile for three of the four anomalous companies. This result indicates that the connectivity with nodes-of-interest participating in triangles provides an informative signal for anomaly detection in the waste management network.

6 Conclusions

This work proposes a Conditional Motif-based Graph Convolutional Network for detecting anomalous activity in waste management networks, by explicitly encoding higher-order connectivity patterns based on other node attributes. Unlike standard GCNs, which rely solely on pairwise relationships, the CM-GCN integrates conditional motif-induced adjacency into its message-passing mechanism. This allows the model to embed structural patterns that are strongly associated with illicit behaviour, that is, interaction with nodes-of-interest (individuals) participating in triangles.

Combined with an LSTM-VAE model, the proposed framework successfully identifies periods in which companies exhibit unusual motif-based connectivity, producing anomaly scores that highlight entities whose behaviour deviates from learned normal patterns. Experimental results on a year of Portuguese waste transport data (e-GAR) show that the CM-GCN-LSTM-VAE substantially outperforms a standard GCN baseline without motif adjacency, which fails to detect any of the known anomalous companies. The method performs marginally better than an LSTM-VAE using a manually engineered triangle-based feature, demonstrating that the CM-GCN is able to learn this structure directly from the graph.

Overall, the findings confirm that (1) introducing the conditional motif adjacency enabled the CM-GCN model to effectively capture suspicious patterns in the network behaviour of the waste operating companies much better than the motif-less GCN, and (2) Attribute-centered triangular motifs provide powerful insights that remain hidden in first-order relationships, thereby contributing to higher detection accuracy.

This approach offers an effective, scalable tool for anomaly detection in waste management, and is currently being integrated in an application, in combination with other approaches, for the practical use of the regulatory authorities of IGAMAOT. This application flags companies with high anomaly scores, to guide and prioritize the routine inspections, and therefore enhance the oversight of waste management operations.

Acknowledgements. This work was co-funded by national funds for Project EnSafe through FCT - Fundação para a Ciência e a Tecnologia, I.P., under the support 2024.07677.IACDC, and by the European Regional Development Fund (ERDF) through the Innovation and Digital Transition Programme (COMPETE 2030) under Portugal 2030 and by National Funds through the FCT - Fundação para a Ciência e a Tecnologia, I.P. (Portuguese Foundation for Science and Technology) within project FOXPM, with reference 14731 (COMPETE2030-FEDER-00923300).

References

1. Bevilacqua, B., et al.: Equivariant subgraph aggregation networks. ArXiv **abs/2110.02910** (2021). <https://api.semanticscholar.org/CorpusID:238407774>
2. Bodnar, C., et al.: Weisfeiler and lehman go cellular: CW networks (2022). <https://arxiv.org/abs/2106.12575>
3. Bodnar, C., et al.: Weisfeiler and lehman go topological: message passing simplicial networks (2021). <https://arxiv.org/abs/2103.03212>
4. Caville, E., Lo, W.W., Layeghy, S., Portmann, M.: Anomal-e: a self-supervised network intrusion detection system based on graph neural networks. *Knowl. Based Syst.* **258**, 110030 (2022)
5. Ding, K., Li, J., Bhanushali, R., Liu, H.: Deep anomaly detection on attributed networks. In: *Proceedings of the 2019 SIAM international conference on data mining*, pp. 594–602. SIAM (2019)
6. Fan, H., Zhang, F., Li, Z.: Anomalydae: dual autoencoder for anomaly detection on attributed networks. In: *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5685–5689. IEEE (2020)
7. e GAR: electronic transport document (2025). <https://apoiosiliamb.apambiente.pt/temas/e-gar>
8. Hamilton, W., Ying, Z., Leskovec, J.: Inductive representation learning on large graphs. *Adv. Neural Inf. Process. Syst.* **30** (2017)
9. IGAMAOT: inspeção-geral da agricultura, do mar, do ambiente e do ordenamento do território (2025). <https://www.igamaot.gov.pt/>
10. Kim, H., Lee, B.S., Shin, W.Y., Lim, S.: Graph anomaly detection with graph neural networks: Current status and challenges. *IEEe Access* **10**, 111820–111829 (2022)

11. Lee, J.B., Rossi, R.A., Kong, X., Kim, S., Koh, E., Rao, A.: Graph convolutional networks with motif-based attention. In: Proceedings of the 28th ACM International Conference On Information And Knowledge Management, pp. 499–508 (2019)
12. Li, S., Zhou, J., Mo, C., Li, J., Tso, G.K., Tian, Y.: Motif-aware temporal GCN for fraud detection in signed cryptocurrency trust networks. arXiv preprint [arXiv:2211.13123](https://arxiv.org/abs/2211.13123) (2022)
13. Li, Y., et al.: Glad: content-aware dynamic graphs for log anomaly detection. In: 2023 IEEE International Conference on Knowledge Graph (ICKG), pp. 9–18. IEEE (2023)
14. Mesgaran, M., Hamza, A.B.: A graph encoder-decoder network for unsupervised anomaly detection. *Neural Comput. Appl.* **35**(32), 23521–23535 (2023)
15. Morris, C., et al.: Weisfeiler and leman go neural: higher-order graph neural networks. *Proc. AAAI Conf. Artif. Intell.* **33**(01), 4602–4609 (2019). <https://doi.org/10.1609/aaai.v33i01.33014602>, <https://ojs.aaai.org/index.php/AAAI/article/view/4384>
16. Oliveira, S., Tabassum, S., Gama, J., Santana, P.: Detecting suspicious activities in waste transport data via temporal and network-aware change detection. In: Machine Learning and Knowledge Discovery in Databases. Workshops Track: European Conference, ECML PKDD 2025, Porto, Portugal, September 15–19. vol. in press. Springer (2025)
17. Shaji, N., Tabassum, S., Ribeiro, R.P., Gama, J., Santana, P., Garcia, A.: Network-based anomaly detection in waste transportation data. In: International Conference on Complex Networks and Their Applications, pp. 273–284. Springer (2024)
18. Tian, Z., Zhuo, M., Liu, L., Chen, J., Zhou, S.: Anomaly detection using spatial and temporal information in multivariate time series. *Sci. Rep.* **13** (2023). <https://api.semanticscholar.org/CorpusID:257560549>
19. Yuan, Z., Shao, M., Yan, Q.: Motif-level anomaly detection in dynamic graphs. *IEEE Trans. Inf. Forensics Secur.* **18**, 2870–2882 (2023)
20. Zhang, Z., Song, L., Bao, E., Lv, X., Wang, X.: Atm-gad: adaptive temporal motif graph anomaly detection for financial transaction networks. arXiv preprint [arXiv:2508.20829](https://arxiv.org/abs/2508.20829) (2025)
21. Zhao, H., et al.: Multivariate time-series anomaly detection via graph attention network. In: 2020 IEEE International Conference On Data Mining (ICDM), pp. 841–850. IEEE (2020)
22. Zheng, L., Li, Z., Li, J., Li, Z., Gao, J.: Addgraph: anomaly detection in dynamic graph using attention-based temporal GCN. In: IJCAI. vol. 3, p. 7 (2019)